

E.1 / PAPER 1 / QUESTION 31

2 May 2023 / TZ1 (file label) / HL Paper 1 / original Q38 / 2223-6507

- 31.** In the Bohr model for hydrogen, the radius of the electron orbit in the $n = 2$ state is four times that of the radius in the $n = 1$ state.

What is $\frac{\text{speed of the electron in the } n = 2 \text{ state}}{\text{speed of the electron in the } n = 1 \text{ state}}$?

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$
- C. 2
- D. 4